

GRAFTON AP, Australia

WMO: 955700

Lat: 29.7583S

Lon: 153.0297E

Elev: 84

StdP: 14.65

Time Zone: 10.00 (AUS)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) 35.8	(c) 38.3	(d) 26.6	(e) 20.6	(f) 63.1	(g) 30.5	(h) 24.8	(i) 57.0	(j) 16.0	(k) 65.5	(l) 14.2	(m) 65.5	(n) 1.4	(o) 190	(p) 0.349

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 1	(b) 18.4	(c) 92.8	(d) 72.5	(e) 89.5	(f) 72.0	(g) 86.6	(h) 71.0	(i) 76.0	(j) 86.6	(k) 74.7	(l) 84.1	(m) 73.5	(n) 82.3	(o) 10.6	(p) 10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.3	124.2	79.0	72.0	118.8	78.0	71.0	114.5	77.2	39.6	87.1	38.3	84.4	37.2	82.6	85.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 16.4	(b) 14.5	(c) 13.2	(e) 30.8	(f) 102.0	(g) 1.5	(h) 5.0	(i) 29.7	(j) 105.6	(k) 28.8	(l) 108.5	(m) 28.0	(n) 111.3	(o) 26.9	(p) 114.9		
			DB													
			WB													
				30.0		79.3		1.4		2.1		29.0		80.8		28.1
																82.0
																27.3
																83.2
																26.3
																84.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	66.2	75.7	75.1	72.7	67.2	61.0	57.0	55.2	57.1	62.4	66.9	71.2	73.4		
	DBStd	8.26	3.90	3.88	3.21	3.52	3.99	4.14	4.19	4.22	4.39	4.70	4.43	4.03		
	HDD50	7	0	0	0	0	0	2	4	1	0	0	0	0		
	HDD65	1087	0	0	0	16	135	241	307	249	101	33	4	1		
	CDD50	5911	796	702	703	516	340	211	164	221	373	523	635	727		
	CDD65	1516	331	283	238	82	10	1	1	4	23	91	189	263		
	CDH74	10824	2535	1993	1321	425	89	4	13	98	340	827	1354	1825		
	CDH80	3282	902	658	298	47	4	1	0	20	80	253	446	573		
(14) Wind	WSAvg	5.3	6.1	6.6	5.5	5.1	3.9	4.1	3.9	4.6	5.2	5.9	6.2	6.3		
(15) Precipitation	PrecAvg	49.4	5.9	6.8	6.5	3.9	3.9	4.1	2.0	2.3	1.9	3.0	4.7	4.6		
	PrecMax	74.2	19.6	17.2	20.2	11.1	17.7	12.6	7.3	9.1	4.3	14.9	8.7	12.7		
	PrecMin	30.1	0.7	1.3	1.5	0.5	0.0	0.9	0.0	0.0	0.2	0.7	0.8	1.8		
	PrecStd	11.3	4.2	4.2	3.9	2.4	4.0	3.3	1.6	2.6	1.1	2.8	2.1	2.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	96.6	96.6	91.5	84.3	80.0	73.7	75.6	81.8	87.4	92.2	95.6	96.1		
		MCWB	74.1	74.9	73.4	68.2	64.3	60.4	59.5	60.0	63.0	67.2	68.7	72.4		
	2%	DB	91.6	90.8	85.8	81.0	76.4	70.8	72.0	76.4	81.4	86.5	89.1	89.3		
		MCWB	74.0	74.7	72.0	67.5	63.2	59.2	57.3	58.5	62.3	66.8	70.5	71.4		
	5%	DB	87.9	86.8	83.1	78.5	73.7	68.9	69.0	72.6	77.6	82.0	84.6	85.7		
		MCWB	73.0	72.6	70.8	66.2	62.0	58.3	56.6	57.1	61.5	65.4	69.3	70.5		
	10%	DB	84.8	83.8	80.6	76.3	71.4	66.9	66.5	69.5	74.3	78.3	81.3	82.7		
		MCWB	72.0	70.9	69.5	65.6	60.3	57.4	55.2	55.6	60.2	64.4	68.2	70.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	77.4	78.6	76.4	71.5	67.6	64.3	65.0	64.5	68.2	72.0	75.5	76.0		
		MCDB	90.9	88.5	83.8	78.4	74.9	68.1	70.5	75.0	78.9	82.7	88.7	88.0		
	2%	WB	75.6	76.3	74.1	69.6	65.8	62.2	61.7	61.7	65.8	69.5	73.0	74.2		
		MCDB	86.4	86.0	81.2	76.5	72.4	67.0	66.6	70.0	75.4	81.0	84.0	84.5		
	5%	WB	74.5	74.6	72.7	68.3	64.1	60.9	59.4	59.7	64.1	67.5	71.3	72.8		
		MCDB	83.8	83.2	79.5	75.4	70.4	65.5	65.3	67.6	72.9	77.3	80.4	82.0		
	10%	WB	73.3	72.9	71.4	66.9	62.4	59.4	57.4	58.0	62.4	66.1	69.8	71.6		
		MCDB	81.8	81.0	77.9	73.9	68.5	64.1	63.4	66.0	70.3	75.1	77.4	79.6		
(35) Mean Daily Temperature Range	MDBR		18.4	17.5	16.9	18.9	22.6	21.3	25.2	27.5	25.1	22.8	20.3	18.8		
	5% DB	MCDBR	24.3	23.5	21.3	22.3	25.1	23.6	28.4	32.5	31.0	30.8	26.8	24.9		
		MCWBR	8.9	8.3	8.6	10.0	12.7	13.0	15.0	16.2	14.0	13.0	10.7	9.6		
	5% WB	MCDBR	20.1	19.7	17.9	18.8	20.2	16.5	20.7	25.2	25.3	24.6	21.7	19.9		
		MCWBR	8.3	8.4	8.5	9.1	11.0	9.8	12.0	13.7	12.9	11.0	9.5	8.5		
(40) Clear-Sky Solar Irradiance	taub		0.387	0.373	0.356	0.337	0.311	0.304	0.298	0.312	0.356	0.391	0.386	0.390		
	taud		2.507	2.560	2.604	2.624	2.656	2.661	2.662	2.600	2.471	2.406	2.473	2.486		
	Ebn at Noon		303	301	297	287	281	276	283	291	291	292	301	303		
	Edh at Noon		36	33	30	27	23	22	23	27	34	38	37	37		
(44) All-Sky Solar Radiation	RadAvg		2064	1870	1576	1345	1098	914	1048	1337	1678	1870	2003	2059		
	RadStd		167	152	100	69	72	57	81	95	120	168	185	190		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (17 neighbors)		+0.72	N/A	N/A	N/A	N/A	N/A	N/A	-101	+301	+162

Nomenclature: See separate page